Evaluating HMC performance on challenging posterior distributions

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Abstract

We evaluate the performance of Hamiltonian Monte Carlo on three notoriously challenging posterior distributions: Hierarchical logistic regression, Neal's Funnel, The Rosenbrock distribution; we do this using a Bayesian framework and the probabilistic programming language Stan, with the goal of contributing to the work of `posteriordb` on benchmarking of inference algorithms. Our results show that HMC performs quite well in these challenging situations.

1. Introduction

Among the existing Markov Chain Monte Carlo (MCMC) sampling algorithms, Hamiltonian Monte Carlo (HMC) has been shown to exhibit superior performance over many others when sampling from target distributions (Neal, 2011, pp. 15-21). When target distributions become complex, such as when the dimensionality of the parameter space increases, HMC succeeds in exploring the target distribution where most other MCMC variants fail (Betancourt, 2017, pp. 15-17).

With that said, there are situations where HMC struggles, for example, when the geometry of a high-dimensional posterior becomes complex, e.g., curved or funnel-shaped. In an attempt to improve existing — and develop new — inference algorithms, the database `posteriordb` (Magnusson et al., 2024) was set up. It contains a collection of pairs of datasets and models that together form posteriors, which can be used to benchmark the performance of inference algorithms, such as HMC.

In this report, we evaluate the performance of HMC against a few posteriors that are known to cause problems for the algorithm. We do this by fitting Stan models to real and simulated data. The models are described in Section 3.

2. Data

The data used will vary depending on the model. For model one (Hierarchical logistic regression), we will use a dataset containing information on whether an individual is a good or bad credit risk, where $Y = 1$ denotes "good credit risk" (UCI Machine Learning, 2025). The response variable is binary, and the number of observations is $N = 522$ after removing missing values. The initial data has 9 predictors; encoding the categorical predictors as dummy variables gives 19 predictors in total. Extending the data to include all two-way interactions results in 146 predictors in total. All numerical predictors are standardized to have a mean of 0 and a standard deviation of 1.

For models two and three, we generate differing datasets using the generated quantities block in Stan (Stan Development Team, 2025, pp. 184-185). We do this by varying elements in the model specification, such as parameters or dimensionality, to create posteriors with varying geometries on which the HMC sampler can be evaluated. See Listings 3 and 5 in Appendix A for the Stan code implementing this data generation. For the second model (Neal's funnel), described in (2), we generate 9 different datasets, varying the dimensionality P of the set of all x-variables $\{X_k\},\ k = 1 \ldots, P$. We use the following values of P: $(1, 3, 9, 20, 50, 75, 100, 150, 200)$. For model three (Hybrid Rosenbrock distribution), described in (3), the variables $\{x_{j,i}\}$ define the dimensionality of the distribution; we will use $n_j = 2$ blocks and $n_i = 3$ variables per block. The parameters μ, a and $b_{j,i} = b$ define the distribution's geometry. We generate 9 different datasets by varying these parameters. Table 1 shows the parameter combinations used to generate the datasets.

Table 1: Parameter combinations used when generating datasets of the Hybrid Rosenbrock distribution. The Stan code implementation can be found in Listing 5 in Appendix A.

	μ	a	b
SET 1	-4	0.05	5
SET 2	0	0.05	5
SET 3	10	0.05	5
SET 4	1	0.5	5
SET 5	1	0.05	5
SET 6	1	0.005	5
SET 7	1	0.05	50
SET 8	1	0.05	0.05
SET 9	1	0.05	0.0005

3. Models

We fit and evaluate three different models. Since we will use different datasets for each model, we will not be able to compare performance between models. For model one, we compare centred and non-centred parametrizations and analyse the estimated parameters. For models two and three, we compare the performance of each model on several varying posterior geometries (described in Section 2).

3.1. Hierarchical logistic regression

The model, described in (Hoffman & Gelman, 2014), involves a binary response; thus, logistic regression is used to model the association between X and Y

$$\Pr(Y_i = 1 \mid \mathbf{X}_i) = \text{Inv-logit}(\alpha_{g_i} + \mathbf{X}_i^\top \boldsymbol{\beta})$$

where α_{g_i} and $\boldsymbol{\beta}$ are the parameters of interest. The `Job` variable will be used as a grouping variable, with four groups $g_i \in \{1, \ldots, 4\}$ treated as nominal ranging from *unskilled* to *highly qualified*. For each group, a group-specific intercept α_{g_i} is estimated. We further assume that observations within each group g, and the parameters (α, β) across the groups, are exchangeable (Bernardo, 1996). Exchangeability implies that de Finetti's representation theorem holds, which in turn implies that the joint distribution of the parameters can be written as

$$p(\alpha_1, \ldots, \alpha_4, \boldsymbol{\beta}) = \\ p(\boldsymbol{\beta} \mid \tau) p(\tau) \int \prod_{g=1}^{4} p(\alpha_g \mid \mu_\alpha, \sigma_\alpha) p(\mu_\alpha) p(\sigma_\alpha) d\mu_\alpha d\sigma_\alpha.$$

This means that the α_gs may be regarded as a random sample from a population distribution of intercepts, with overall mean μ_α and overall standard deviation σ_α. The assumption of exchangeability allows us to utilize hierarchical modelling. The slope parameters $\boldsymbol{\beta} \in \mathbb{R}^P$ are the same for each observation. We use a non-centred parametrization on both α_{g_i} and $\boldsymbol{\beta}$; the model is defined as

$$y_i \mid x, \alpha_{g_i}, \beta \sim \text{Bernoulli}(\text{logit}^{-1}(\alpha_{g_i} + \mathbf{X}_i^\top \boldsymbol{\beta}))$$

$$\begin{aligned} \tilde{\boldsymbol{\beta}} &\sim \text{Normal}(0, 1) \\ \sigma_\beta^2 &\sim \text{Exponential}(0.01) \\ \boldsymbol{\beta} &= \tilde{\boldsymbol{\beta}} \sqrt{\sigma_\beta^2} \end{aligned} \tag{1}$$

$$\begin{aligned} \tilde{\alpha}_g &\sim \text{Normal}(0, 1) \\ \sigma_\alpha^2 &\sim \text{Exponential}(0.01) \\ \alpha_g &= \tilde{\alpha}_g \sqrt{\sigma_\alpha^2}. \end{aligned}$$

3.2. Neal's funnel

Neal's funnel (Neal, 2003) is a joint distribution of the variables $(\nu, x_1, \ldots, x_P)$, where the number of variables in $\{x_k\}$ defines the dimensionality of the model. The parameter of interest ν controls the scale of the x-variables; when ν is negative, the variance of $\mathbf{x} = (x_1, \ldots, x_k)$ becomes small, but when ν is positive, the variance becomes large – together, this creates a funnel geometry. The model is defined as

$$\begin{aligned} \nu &\sim \text{Normal}(0, 3^2) \\ x_k \mid \nu &\sim \text{Normal}(0, e^\nu), \qquad k = 1, \ldots, P. \end{aligned} \tag{2}$$

3.3. Hybrid Rosenbrock distribution

The Hybrid Rosenbrock distribution (Pagani et al., 2020) is an n-dimensional distribution for which the kernel of the distribution can be written as

$$p(\mathbf{x}) \propto \exp\left[-a(x_1 - \mu)^2 - \sum_{j=1}^{n_j} \sum_{i=2}^{n_i} b_{j,i}(x_{j,i} - x_{j,i-1}^2)^2\right] \tag{3}$$

where $\mu, a, b_{j,i}$ are the parameters of interest. $\mu, x_{j,i} \in \mathbb{R}$, and $a, b_{j,i} \in \mathbb{R}^+$. The variables $x_{j,i}$ follow a dependent block structure, where $j = 1, \ldots, n_j$ represents the block, and $i = 2, \ldots, n_i$ represents the variable inside a block. x_1 is shared for each block; each block has $n_i - 1$ unique variables (excluding the shared x_1), and the total variable dimension is $n = (n_i - 1)n_j + 1$. It is possible to see that (3) is made up of normal kernels, where every variable except the root variable x_1 depends on the previous variable in the block j. This dependency structure induces a curved posterior geometry that is challenging for sampling algorithms.

4. Results

In this section we present results for each of the three models respectively. Posterior quantiles and Monte Carlo standard errors can be found in Appendix B.

4.1. Hierarchical logistic regression results

Figure 1 plots the joint posteriors of the group intercept parameters and the population variance σ_α^2 side by side, comparing models with non-centred parametrization (NCP) and centred parametrization (CP). We observe more divergent transitions with the CP. Further, the NCP allows the algorithm to draw samples with lower values of σ_α, tracing out more of the true funnel distribution.

Figure 2 shows the median odds ratio estimate with 95% credible intervals for the slope parameters with the largest

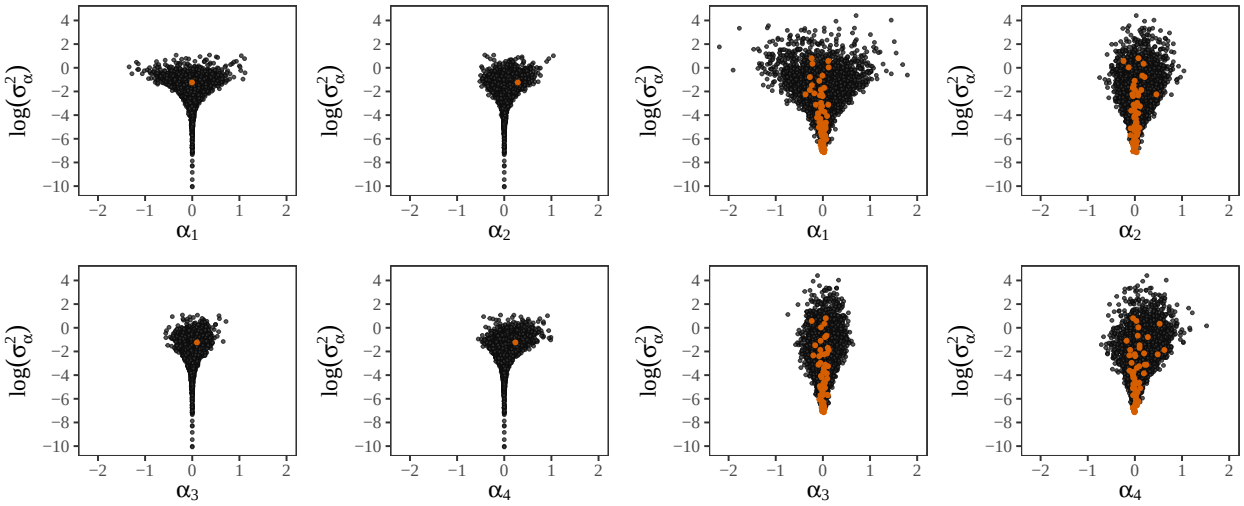

Figure 1: Funnel-shaped marginal distributions for the Hierarchical logistic regression model. **(four left plots):** Non-centred parametrization. There was one divergent transition (in orange). **(four right plots):** centred parametrization. There were 38 divergent transitions (in orange); all four chains had low EFMI; the bulk ESS was low. Intercepts for the four different groups (log-odds scale) on the horizontal axis, and the variance hyperparameter (log scale) is on the vertical axis.

absolute deviation from odds ratio $= 1$. $\beta_3 = 0.838$ corresponding to $x_3 =$ `Duration` (in months) shows the largest absolute deviation from an odds ratio of 1. We interpret this as: an increase in the duration of the loan by one standard deviation – holding all other variables unchanged – leads to a decrease in the odds of being a good credit risk by 16.2%.

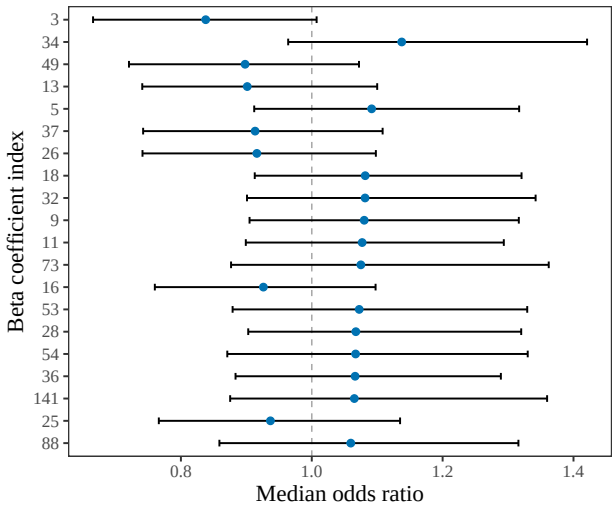

Figure 2: NCP. Median odds ratios with 95% credible intervals. The top 20 beta parameters in terms of distance from odds ratio = 1 (dashed line) are displayed.

For the NCP, all $\hat{R}$-values are below 1.01. For the CP, two values of $\hat{R}$ exceed this threshold: `lp`: 1.024 and `sigma2_beta`: 1.022.

4.2. Neal's funnel results

Table 2 presents results for using HMC to sample from Neal's funnel. The model in (2) was implemented in Stan using code in Listing 4 in Appendix A.

Table 2: Results for Neal's funnel. Sampling using HMC was done on nine datasets with varying x-dimension and varying true ν.

DIM(x)	TRUE ν	MEAN$_\nu$	SD$_\nu$	RMSE
1	1.438	1.419	0.014	0.024
3	1.087	1.088	0.008	0.008
9	-5.987	-5.984	0.005	0.005
20	3.2714	3.2732	0.0032	0.0036
50	-2.2615	-2.2667	0.0020	0.0056
75	0.3749	0.3742	0.0016	0.0017
100	-0.7254	-0.7258	0.0014	0.0015
150	-0.3074	-0.3050	0.0012	0.0026
200	1.7661	1.7674	0.0010	0.0016

Table 2 shows that the posterior mean is close to the true value of ν in all datasets. With a few exceptions, the results indicate a trend of a decrease in posterior standard deviation and RMSE as the dimensionality increases. Differences in RMSE between datasets are generally quite small, except between $P = 1$ and $P = 3$, where RMSE decreases threefold. The results indicate that HMC performance increases when dimensionality increases. This might be explained by what

Table 3: Results for the Hybrid Rosenbrock distribution. Sampling using HMC was done on nine datasets with varying parameters (μ, a, b).

	μ				a				b			
	TRUE	**MEAN**	**SD**	**RMSE**	**TRUE**	**MEAN**	**SD**	**RMSE**	**TRUE**	**MEAN**	**SD**	**RMSE**
SET 1	-4	-3.964	0.031	0.04731	0.05	0.0506	0.0007	0.00096	5	5.000	0.035	0.03542
SET 2	0	0.037	0.031	0.04809	0.05	0.0506	0.0007	0.00096	5	5.000	0.035	0.03531
SET 3	10	10.036	0.031	0.04794	0.05	0.0506	0.0007	0.00941	5	5.000	0.035	0.03541
SET 4	1	1.011	0.010	0.01514	0.5	0.506	0.0070	0.00981	5	5.000	0.036	0.03607
SET 5	1	1.037	0.032	0.04861	0.05	0.0506	0.0007	0.00096	5	5.000	0.035	0.03534
SET 6	1	1.12	0.1	0.15307	0.005	0.00506	0.00007	0.00009	5	5.000	0.036	0.03584
SET 7	1	1.036	0.032	0.04783	0.05	0.0507	0.0007	0.00099	50	50.00	0.35	0.34977
SET 8	1	1.036	0.032	0.04799	0.05	0.0506	0.0007	0.00096	0.05	0.05000	0.00036	0.00036
SET 9	1	1.037	0.031	0.04799	0.05	0.0506	0.0007	0.00009	0.00005	0.00005	0.000003	0.000003

Betancourt (2017) calls the concentration of measure phenomenon. This means that in higher dimensions, the region of the joint posterior from which impactful contributions to expectations come from (what he calls the typical set), becomes increasingly tight and narrow. Somewhat counter-intuitively, this may make sampling easier for HMC.

$\hat{R} \geq 1.01$ for two parameters. $\hat{R} = 1.01$ for ν when $P = 100$. For the log posterior density `lp`, $\hat{R} = 1.01$ when $P = 200$. For all other datasets, the values are equal to one.

4.3. Hybrid Rosenbrock distribution results

Table 3 presents results for using HMC to sample from the Hybrid Rosenbrock distribution. It shows true values, and posterior quantities for the sampled parameters of interest (μ, a, b). The model in (3) was implemented in Stan using code in Listing 6 in Appendix A. Table 3 shows that the posterior means for all three variables are close to their true values, indicating that HMC is able to perform well on this posterior. RMSE is quite small for all parameters across all datasets. The exception is for μ in Set 6, which is roughly three times larger than for the other sets. $\hat{R} < 1.01$ for all parameters.

5. Conclusions

We set out to evaluate how well Hamiltonian Monte Carlo (HMC) performs on three posteriors that are known to be difficult for the sampling algorithm due to a curved or funnel-shaped posterior geometry. We implemented the posterior sampling in Stan; the results provide evidence that HMC performs reasonably well even in our challenging settings.

The first posterior consisted is a hierarchical logistic regression model fitted to data on bank customers' credit risk. HMC performed poorly using a centred-parametrization, resulting in many divergent transitions; using a non-centred parametrization alleviated this problem.

For the second posterior — the Neal's famous funnel — we generated nine datasets with varying variance hyperparameter ν. The posterior mean of ν was close to the true value across all datasets, while RMSE tended to be higher for lower dimensionality of x. Thus, HMC is able to sample adequately from this posterior, but struggles more as dimensionality decreases.

For the Hybrid Rosenbrock distribution we also generated nine datasets, varying the parameters (μ, a, b). The posterior means were all close the their respective true values, and RMSE values were low.

$\hat{R}$ values were generally below 1.01 for all models, with a few exceptions where the values for some parameters were slightly higher than this boundary.

6. Acknowledgements

We hereby grant our consent for the utilization of this project report as a reference material within the context of future editions of the course.

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A. Stan code

Listing 1: Hierarchical logistic regression (CP)

```
data {
  int N;
  int P;
  int<lower=1> N_g;  // Number of levels in covariate g
  array[N] int<lower=1,upper=N_g> g_id;  // Mapping from obs to level in g
  array[N] int<lower=0,upper=1> y;
  matrix[N,P] X;
}

parameters {
  // Shared slope vector for all obs
  vector[P] beta;
  real<lower=0> sigma2_beta;

  // Group intercepts
  vector[N_g] alpha_g;
  real<lower=0> sigma2_alpha_g;
}

model {
  // Log-likelihood
  target += bernoulli_logit_lpmf(y | alpha_g[g_id] + X * beta);

  // Log-priors
  target += normal_lpdf(beta | 0, sqrt(sigma2_beta));
  target += exponential_lpdf(sigma2_beta | 0.01);

  target += normal_lpdf(alpha_g | 0, sqrt(sigma2_alpha_g));
  target += exponential_lpdf(sigma2_alpha_g | 0.01);
}
```

Listing 2: Hierarchical logistic regression (NCP)

```
data {
  int N;
  int P;
  int<lower=1> N_g;  // Number of levels in covariate g
  array[N] int<lower=1,upper=N_g> g_id;  // Mapping from obs to level in g
  array[N] int<lower=0,upper=1> y;
  matrix[N,P] X;
}

parameters {
  // Shared slope vector for all obs
  vector[P] beta_raw;
  real<lower=0> sigma2_beta;

  // Group intercepts
  vector[N_g] alpha_g_raw;
  real<lower=0> sigma2_alpha_g;
}

transformed parameters {
  vector[P] beta = beta_raw
                 * sigma2_beta;

  vector[N_g] alpha_g = alpha_g_raw
                      * sigma2_alpha_g;
}

model {
  // Log-likelihood
  target += bernoulli_logit_lpmf(y | alpha_g[g_id] + X * beta);

  // Log-priors
  target += normal_lpdf(beta_raw | 0, 1);
  target += exponential_lpdf(sigma2_beta | 0.01);

  target += normal_lpdf(alpha_g_raw | 0, 1);
  target += exponential_lpdf(sigma2_alpha_g | 0.01);
}
```

Listing 3: Neal's Funnel (generate data)

```
data {
  int<lower=1> N;  // Number of observations to generate
  int<lower=1> P;  // Number of x-variables to generate
}

generated quantities {
  matrix[N, P] X;
  real v = normal_rng(0, 3);

  for(n in 1:N) {
    for(p in 1:P) {
      X[n, p] = normal_rng(0, sqrt(exp(v)));
    }
  }
}
```

Listing 4: Neal's Funnel (sampling)

```
data {
  int N;  // Number of observations
  int P;  // Number x-variables
  matrix[N, P] X;
}

parameters {
  real v;
}

model {
  // Log likelihood
  target += normal_lpdf(to_vector(X) | 0, sqrt(exp(v)));

  // Log priors
  target += normal_lpdf(v | 0, 3);
}
```

Listing 5: Hybrid Rosenbrock Distribution (generate data)

```
data {
  int<lower=1> N;  // Number of samples
  int<lower=1> ni;  // Index of variable "inside block"
  int<lower=1> nj;  // Index of "block"
  real mu;
  real<lower=0> a;  // sd-factor for root x1
  real<lower=0> b;  // Shared sd-factor for following variables
}

transformed data {
  int n_dim = (ni-1)*nj+1;  // Total dimension
  real sd1 = 1/sqrt(2*a);  // sd root variable
  real sd_b = 1/sqrt(2*b);  // sd of variables (excl. root)
}

generated quantities {
  matrix[N, n_dim] X;

  for(k in 1:N) {
    int idx = 1;

    // Root variable
    X[k, idx] = normal_rng(mu, sd1);
    real x1 = X[k, idx];
    idx += 1;

    // Blocks
    for(j in 1:nj) {
      real x_prev = x1;  // Reset x_prev for each block

      // Variables in block
      for(i in 2:ni) {
        real x_new = normal_rng(square(x_prev), sd_b);
        X[k, idx] = x_new;
        x_prev = x_new;
        idx += 1;
      }
    }
  }
}
```

Listing 6: Hybrid Rosenbrock Distribution (sampling)

```
data  {
  int N;  // Number of samples
  int<lower=2> ni;  // Index of variable "inside block"
  int<lower=1> nj;  // Index of "block"
  int<lower=1> n_dim;  // Total dimension
  matrix[N, n_dim] X;  // Observed (simulated) data
}

parameters {
  real mu; // Mean of root variable
  real<lower=0> a;  // Variance of "root variable normal distribution"
  real<lower=0> b;  // Variance of "following variables' normal distribution"
}

model {
  // Log Likelihood
  for(n in 1:N) {
    // Root variable x_[j,1]
    target += normal_lpdf(X[n, 1] | mu, inv_sqrt(2 * a));

    // For each block j, the first variable depends on x1.
    for(j in 1:nj) {
      int idx_first = 2 + (j-1) * (ni-1);  // index of x_[j,2]
      target += normal_lpdf(X[n, idx_first] | square(X[n, 1]), inv_sqrt(2 * b));

      // Then each variable depends on *previous variable in block* squared.
      for(i in 3:ni) {
        int idx_cur = idx_first + (i-2);  // index of x_[j,i]
        int idx_prev = idx_cur - 1;  // index of x_[j,i-1]
        target += normal_lpdf(X[n, idx_cur] | square(X[n, idx_prev]), inv_sqrt(2 * b));
      }
    }
  }

  // Log Priors
  target += normal_lpdf(mu | 0, 10);
  target += student_t_lpdf(a | 3, 0, 2.5);
  target += student_t_lpdf(b | 3, 0, 2.5);
}
```

B. Quantiles and Monte Carlo standard errors

Table 4: Quantiles and Monte Carlo standard errors for the parameters α_g, and the β-parameters shown in Figure 2 in the Hierarchical logistic regression model.

PARAMETER	MCSE (MEAN)	Q_{05}	MCSE (Q_{05})	Q_{95}	MCSE (Q_{95})
ALPHA$_g$[1]	0.00639368	-0.61655773	0.01904682	0.54244094	0.01848294
ALPHA$_g$[2]	0.00409066	-0.23450805	0.00993938	0.45314175	0.01593960
ALPHA$_g$[3]	0.00315734	-0.24109502	0.00611530	0.30680666	0.00814477
ALPHA$_g$[4]	0.00441717	-0.20669632	0.00990219	0.56424969	0.01003506
BETA[11]	0.00136128	-0.08071269	0.00581026	0.23145472	0.00582698
BETA[13]	0.00150972	-0.28099066	0.00523160	0.04817880	0.00429311
BETA[141]	0.00249901	-0.11620333	0.00603351	0.29774489	0.00752571
BETA[16]	0.00173841	-0.24129378	0.00396720	0.05774951	0.00342051
BETA[18]	0.00182508	-0.06752052	0.00381860	0.24817661	0.00385928
BETA[25]	0.00141222	-0.23559764	0.00343163	0.09878511	0.00465597
BETA[26]	0.00175191	-0.27308395	0.00469322	0.05652935	0.00384781
BETA[28]	0.00199807	-0.07407317	0.00309940	0.23890501	0.00475403
BETA[32]	0.00157792	-0.07771140	0.00386004	0.26381661	0.00471931
BETA[34]	0.00239599	-0.01695867	0.00293662	0.32276515	0.00631154
BETA[36]	0.00142297	-0.09357114	0.00502673	0.23194646	0.00461766
BETA[37]	0.00150557	-0.27591094	0.00751724	0.08483893	0.00371986
BETA[3]	0.00197367	-0.37807004	0.00705993	-0.02444030	0.00418434
BETA[49]	0.00275449	-0.29469212	0.00652744	0.03729899	0.00367530
BETA[53]	0.00184591	-0.12371325	0.00536967	0.26017775	0.00896365
BETA[54]	0.00172448	-0.12877101	0.00534654	0.25032945	0.00487953
BETA[5]	0.00165452	-0.05091016	0.00301260	0.26249567	0.00673807
BETA[73]	0.00187395	-0.10450330	0.00320596	0.26608945	0.00535167
BETA[88]	0.00146137	-0.11518777	0.00586752	0.24183088	0.00455546
BETA[9]	0.00150318	-0.07166483	0.00392239	0.24729347	0.00509250

Table 5: Quantiles and Monte Carlo standard errors for the parameter ν in the Neal's funnel model.

DIM(x)	MCSE (MEAN)	Q_{05}	MCSE (Q_{05})	Q_{95}	MCSE (Q_{95})
1	0.00039161	1.39643200	0.00088755	1.44109957	0.00075334
3	0.00024489	1.07398804	0.00061144	1.10086539	0.00054823
9	0.00014256	-5.99171446	0.00039742	-5.97666805	0.00022422
20	0.00009588	3.26789426	0.00019619	3.27854121	0.00021709
50	0.00006330	-2.27012166	0.00016121	-2.26344009	0.00011192
75	0.00005235	0.37158174	0.00012216	0.37683578	0.00009752
100	0.00004321	-0.72809547	0.00008195	-0.72354835	0.00011851
150	0.00004046	-0.30705105	0.00012257	-0.30315909	0.00005982
200	0.00002994	1.76572260	0.00005367	1.76891208	0.00005101

Table 6: Quantiles and Monte Carlo standard errors for the parameter μ in the Hybrid Rosenbrock model.

SET	MCSE (MEAN)	Q_{05}	MCSE (Q_{05})	Q_{95}	MCSE (Q_{95})
1	0.00047743	-4.01517060	0.00103372	-3.91356797	0.00107900
2	0.00050632	-0.01359080	0.00121287	0.08894297	0.00146608
3	0.00050370	9.98527599	0.00097503	10.08765954	0.00089538
4	0.00015646	0.99520477	0.00038232	1.02780063	0.00044481
5	0.00053774	0.98426419	0.00119912	1.08847045	0.00132490
6	0.00159612	0.95356335	0.00304616	1.27818913	0.00373589
7	0.00054631	0.98450068	0.00154541	1.08702096	0.00119715
8	0.00052604	0.98354064	0.00163149	1.08862564	0.00179650
9	0.00053467	0.98668528	0.00127411	1.08681112	0.00076791

Table 7: Quantiles and Monte Carlo standard errors for the parameter a in the Hybrid Rosenbrock model.

SET	MCSE (MEAN)	Q_{05}	MCSE (Q_{05})	Q_{95}	MCSE (Q_{95})
1	0.00001130	0.04944992	0.00002894	0.05179992	0.00002558
2	0.00001105	0.04944354	0.00002798	0.05184156	0.00003283
3	0.00001066	0.04945909	0.00002716	0.05179443	0.00002160
4	0.00013294	0.49417396	0.00028315	0.51865156	0.00024037
5	0.00001081	0.04949921	0.00002600	0.05185267	0.00002931
6	0.00000118	0.00494692	0.00000257	0.00518363	0.00000396
7	0.00001146	0.04945705	0.00003100	0.05187042	0.00003348
8	0.00001078	0.04945733	0.00003148	0.05180346	0.00002493
9	0.00001123	0.04947387	0.00002423	0.05180518	0.00002324

Table 8: Quantiles and Monte Carlo standard errors for the parameter b in the Hybrid Rosenbrock model.

SET	MCSE (MEAN)	Q_{05}	MCSE (Q_{05})	Q_{95}	MCSE (Q_{95})
1	0.00052824	4.94209872	0.00112583	5.05849088	0.00145050
2	0.00052136	4.94121670	0.00134513	5.05922615	0.00124640
3	0.00052945	4.94274463	0.00125001	5.05848567	0.00193651
4	0.00053292	4.94188153	0.00124046	5.05922424	0.00130925
5	0.00046995	4.94271060	0.00139732	5.05743270	0.00099929
6	0.00056168	4.93972676	0.00158858	5.05827157	0.00137752
7	0.00482048	49.41570633	0.01173458	50.56923443	0.01843154
8	0.00000506	0.04940594	0.00001158	0.05059918	0.00001337
9	0.00000005	0.00049439	0.00000015	0.00050568	0.00000015